DNS (DOMAIN NAME SYSTEM)

Domain Name System (DNS) is a network service that translates human-readable domain names, such as www.example.com, into the IP addresses that computers use to communicate. Without DNS, users would have to remember the numerical IP address of every website or server they want to access.
When a user enters a domain name, the client sends a query to a DNS server. The server looks up the name in its records and returns the matching IP address. If the server does not have the record, it can forward the query to other DNS servers. DNS uses **UDP** port 53 for most queries and **TCP** port 53 for larger responses and zone transfers.
DNS is an essential part of the Internet and is used in homes, schools, offices, and data centers for accessing websites, email, and network services.

Advantages of DNS

- Allows users to access websites and servers using easy-to-remember names instead of IP addresses.
- Makes network management easier, because a server's IP address can change without users needing to know.
- Provides a distributed and scalable naming system that can handle a very large number of requests.
- Improves speed through caching, since frequently used records are stored and reused.
- Supports multiple record types, such as A, CNAME, MX, and NS, for different services.
- Allows several names to point to the same server, and one name to point to several servers.

Disadvantages of DNS

- The DNS server is a critical component, so if it fails, users cannot access services by name.
- Vulnerable to attacks such as DNS spoofing, cache poisoning, and denial-of-service attacks.
- Incorrect or outdated records can send users to the wrong server or cause connection failures.
- Requires proper configuration and regular maintenance of records.
- Changes to records may take time to spread across the network because of caching.
- A slow or overloaded DNS server can delay access to every service in the network.

Simulation / Design in Cisco Packet Tracer

This section presents the **DNS** network designed and simulated in Cisco Packet Tracer. The network consists of a single server (Server0) and multiple PCs connected to a central switch. Server0 performs three roles: it acts as the DHCP server that automatically assigns IP addresses to the PCs, as the DNS server that resolves the domain name to an IP address, and as the web server that hosts the HTTP/HTTPS page (index.html). The PCs obtain their IP configuration automatically through DHCP and use the DNS server to access the web page by domain name, demonstrating the working of DNS.

Network Diagram

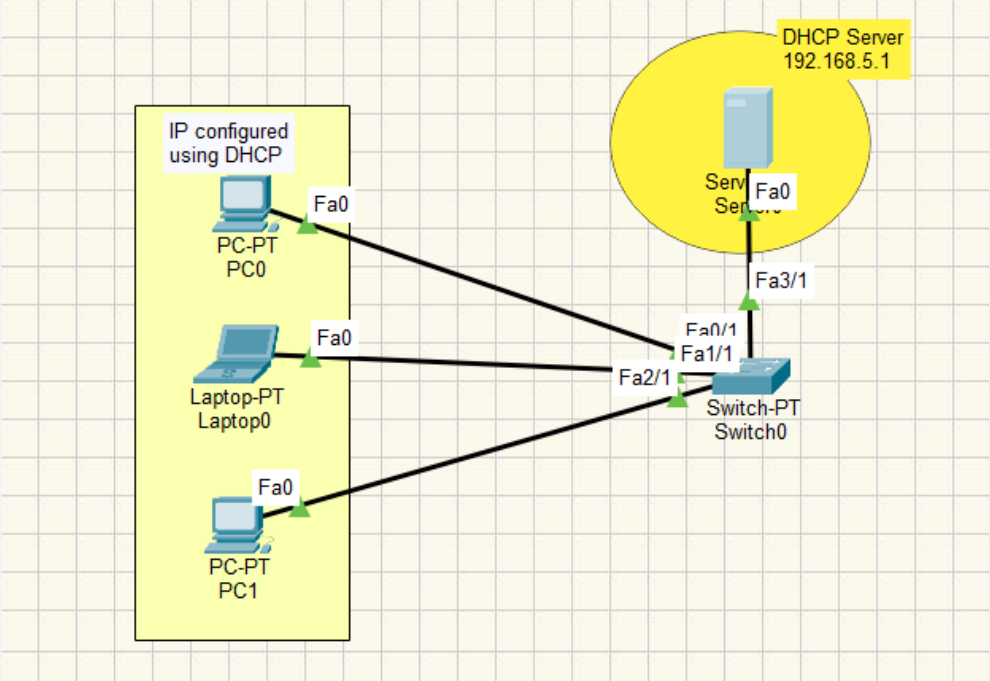


Figure 1: DNS network layout in Cisco Packet Tracer

Device Configuration

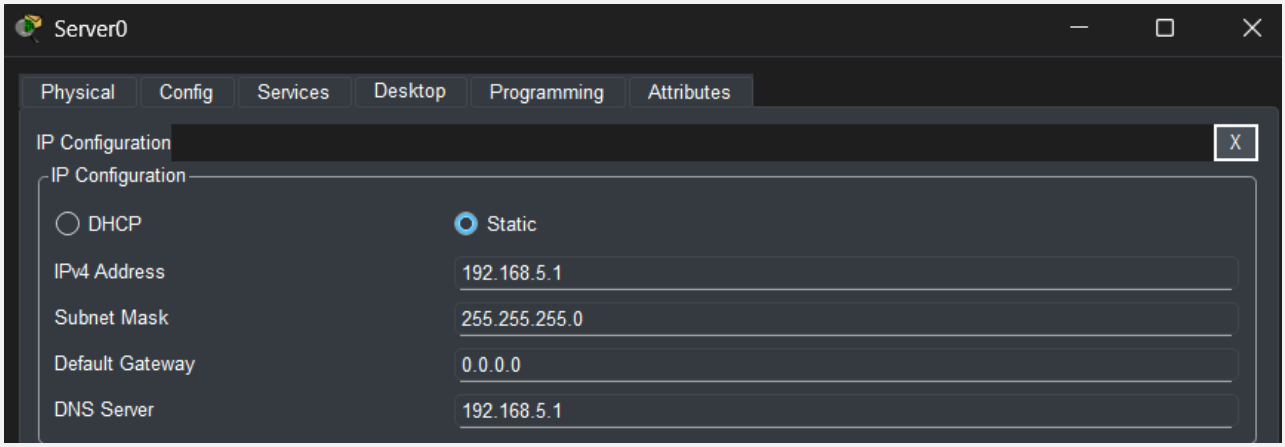


Figure 2.1: Static IP configuration of Server0 (192.168.5.1)

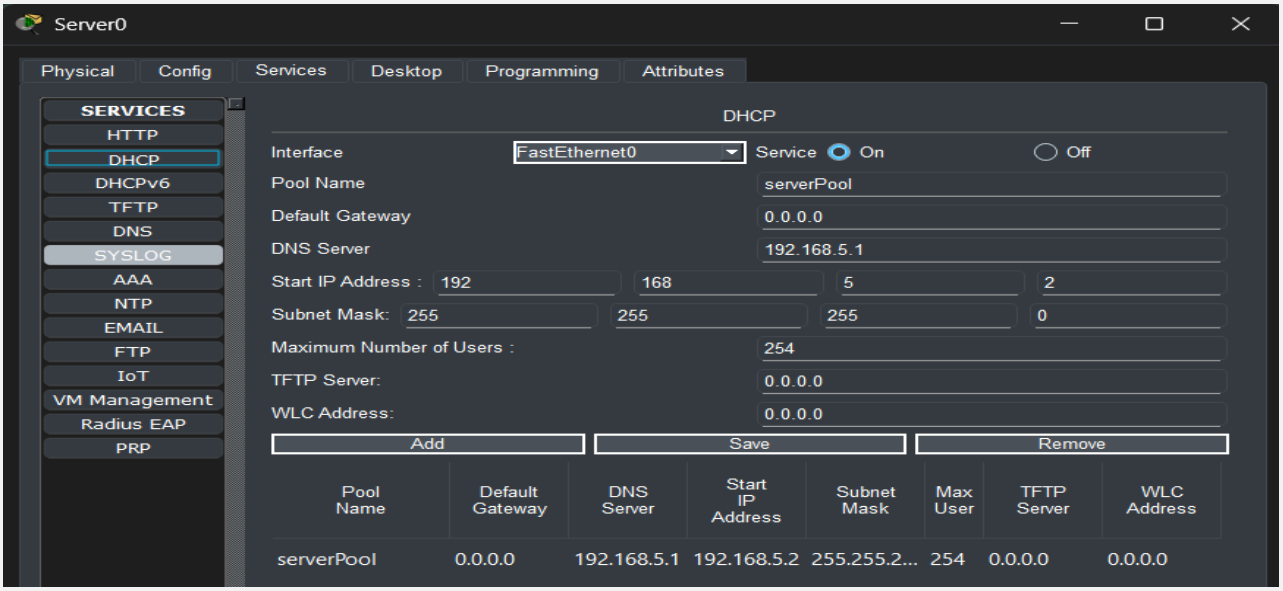


Figure 2.2: DHCP service configuration window of Server0 (serverPool)

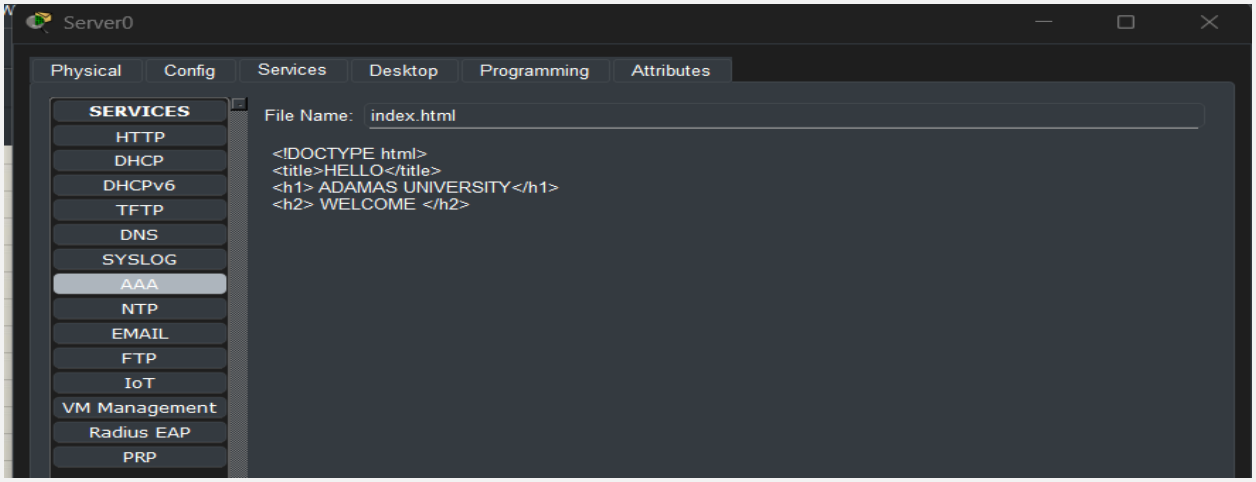


Figure 2.3: HTTP/HTTPS service configuration with the edited index.html file

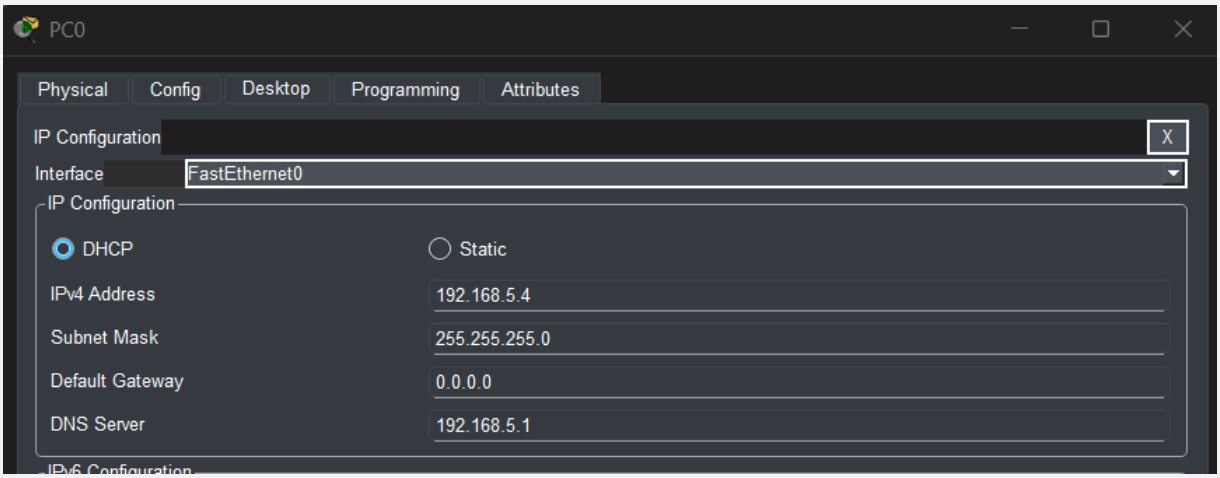


Figure 2.4: IP address automatically assigned to PC0

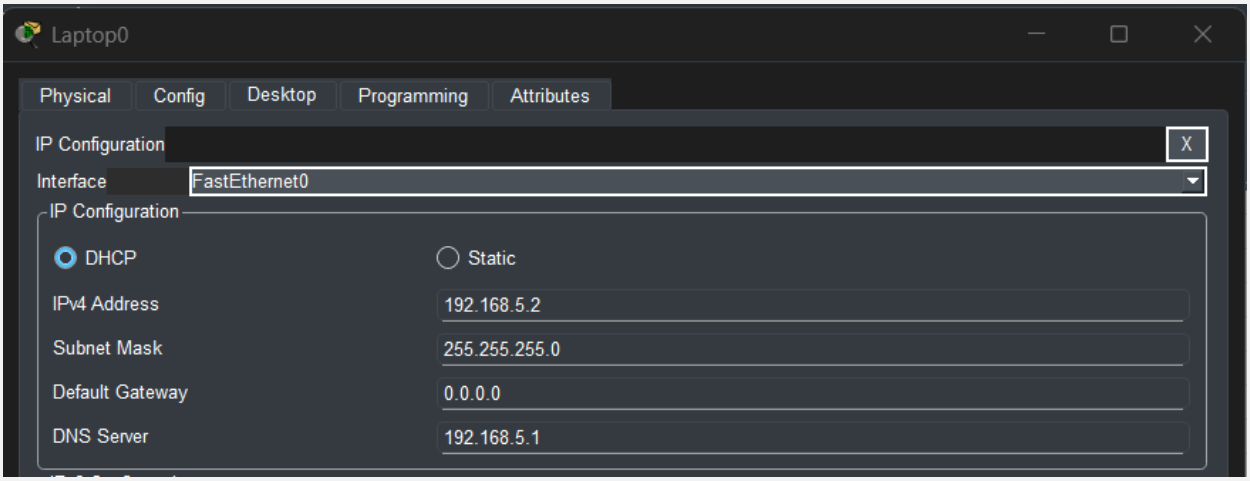


Figure 2.5: IP address automatically assigned to Laptop0

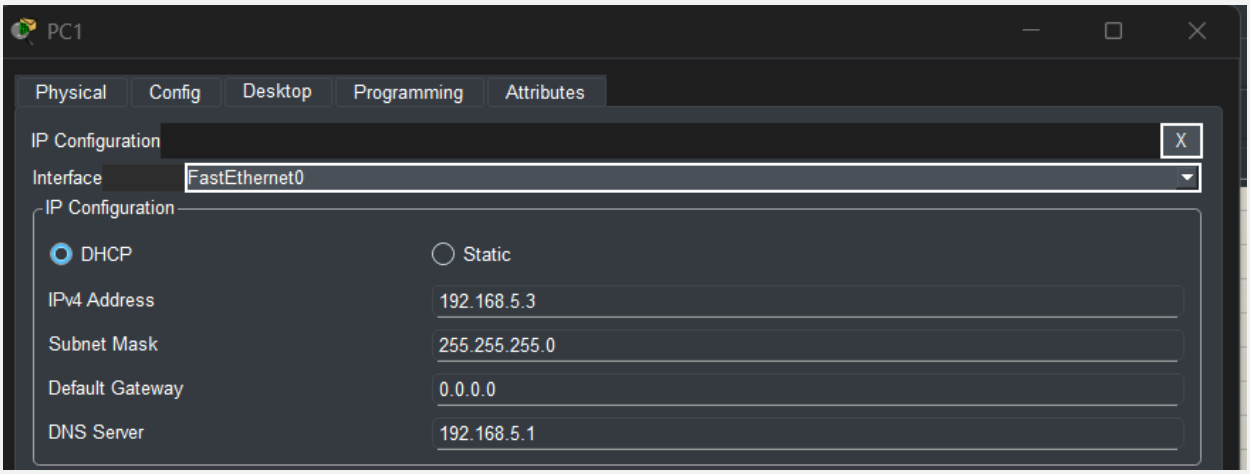


Figure 2.6: IP address automatically assigned to PC1

Connectivity Test (DNS/ Simulation Result)

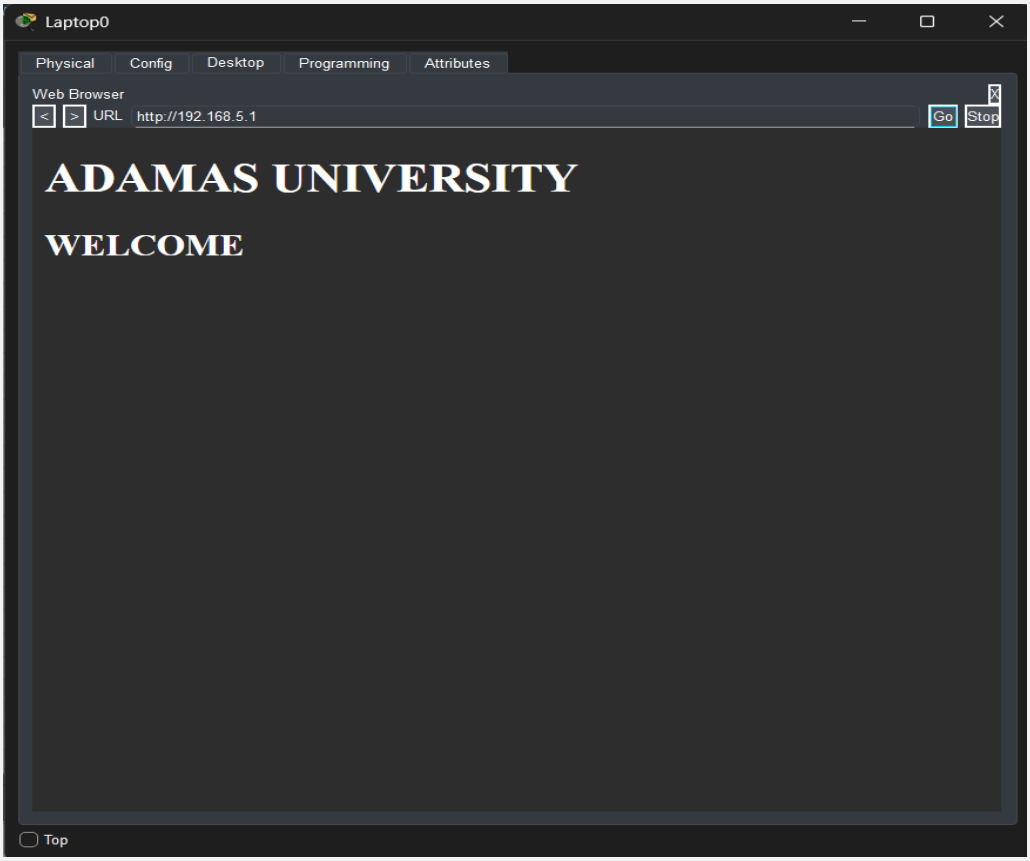


Figure 3.1: Successful access to the web page using the IP address 192.168.5.1 in the browser

Process of Designing the Network in Packet Tracer

- Open Cisco Packet Tracer and create a new blank project to begin designing the DNS network.
- Select a suitable switch from the Network Devices category and place it at the center of the workspace. The switch acts as the central networking device.
- From the End Devices category, place one Server-PT (Server0) and the required number of PCs around the switch.
- Select Copper Straight-Through cable from the Connections category and connect Server0 and each PC individually to the switch. Wait until the link lights turn green, which shows that the physical connections are active.
- Configure Server0 with a static IP address by opening Server0 → Desktop → IP Configuration and selecting Static. The server must keep a fixed address because the PCs depend on it for DHCP, DNS, and web services.

Field	Value
IPv4 Address	192.168.5.1
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	192.168.5.1

- The default gateway is left as 0.0.0.0 because all devices are in the same network and no router is used. The DNS server field points to Server0 itself, since it acts as the DNS server.
- Configure the DHCP service by opening Server0 → Services → DHCP and setting the service to On. Edit the default pool (serverPool) with the following details and click Save.

Field	Value
Pool Name	serverPool
Default Gateway	0.0.0.0
DNS Server	192.168.5.1
Start IP Address	192.168.5.2
Subnet Mask	255.255.255.0
Maximum Number of Users	254

- The pool starts from 192.168.5.2, so the server's own address (192.168.5.1) is never handed out to a client. The DNS server address in the pool ensures that every PC automatically receives 192.168.5.1 as its DNS server.
- Configure the web server by opening Server0 → Services → HTTP and making sure the HTTP and HTTPS services are On. Delete all the default files except index.html, and then edit index.html with the following content.

File Name	index.html
Content	<!DOCTYPE html> <title>HELLO</title> ADAMAS UNIVERSITY WELCOME

- Save the file so that the web server displays this page to every client that connects to it.

- Configure the DNS service by opening Server0 → Services → DNS and turning the DNS service On.
- Add a record for the web server. Enter the name (for example, cisco.com), select the type A Record, and enter the address 192.168.5.1. Click Add to save the record. The address points to Server0, because it hosts the web page.
- Make the PCs request an address. On each PC, open Desktop → IP Configuration and select the DHCP option instead of Static. The message "DHCP request sent" appears briefly, and the PC then displays its newly assigned IP address, subnet mask, and DNS server (192.168.5.1).
- Verify the assigned configuration by opening the Command Prompt of a PC and entering the command ipconfig /all. The output shows the IP address received from the pool (starting from 192.168.5.2), the subnet mask, and the DNS server address.
- Verify the web server using the web browser. Open PC0 → Desktop → Web Browser, enter the IP address of the server (192.168.5.1), and click Go. The web page created in index.html is displayed, confirming that the web server is working.
- Test name resolution by entering the domain name (cisco.com) in the web browser instead of the IP address. The same web page is displayed, which confirms that the DNS server has successfully translated the domain name into the IP address of the web server.
- The domain name can also be tested from the Command Prompt using ping cisco.com or nslookup cisco.com. The name is resolved to 192.168.5.1, and the replies confirm that DNS is working correctly.
- The network can also be tested in Simulation Mode by filtering DNS, DHCP, and HTTP to observe the packets passing through the switch.
- Screenshots of the topology, the server's static IP configuration, the DHCP pool, the HTTP page, the DNS record, the PCs' DHCP configuration, and the successful browser results are captured as evidence of the completed setup.
- Finally, the completed Packet Tracer project is saved as a .pkt file, along with the required documentation and screenshots for submission.

Conclusion

The **DNS** network was successfully designed and configured in Cisco Packet Tracer, demonstrating how domain names are translated into IP addresses. The experiment involved configuring Server0 with a static IP address and running DHCP, DNS, and HTTP/HTTPS services on it. The PCs obtained their IP addresses and DNS server address automatically through DHCP, and the DNS record allowed them to open the web page using a domain name instead of an IP address. Name resolution was verified using the web browser, the ping and nslookup commands, and Simulation Mode. This experiment reinforced the practical understanding of DNS, DHCP, web hosting, and client-server communication in a network.